7 December 1971
SUPERSEDING
MIL-C-43532
21 August 1967

MILITARY SPECIFICATION

CONVEYOR, LAUNDRY, STORAGE BINS, ELECTRIC POWERED, AUTO SELECTIVE CONTROL

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers electric powered conveyors for laundry bundle storage bins (see 6.1).

1.2 Classification.- The laundry bundle conveyors shall be of the following size storage areas as specified (see 6.2).

Size A - 175 cubic feet
Size B - 300 cubic feet
Size C - 365 cubic feet
Size D - 480 cubic feet
Size E - As specified*

* NOTE: Users should specify standard size conveyors only, unless existing operating areas necessitate a change.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

CC-M-636 - Motors, Alternating-Current (Fractional Horsepower).
CC-M-641 - Motors, Alternating-Current (Integral Horsepower, 200 HP and Smaller).

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MILITARY

MIL-L-3153 - Laundry and Dry Cleaning Machinery and Equipment (For Fixed Installations), Preparation for Delivery of.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130 - Identification Marking of U.S. Military Property.
MIL-STD-461 - Electromagnetic Interference Characteristics, Requirements for Equipment.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

National Electrical Manufacturers Association (NEMA) Standards Publication

ICS-1970 - Industrial Controls and Systems (1970)

(Application for copies should be addressed to the National Electrical Manufacturers Association, 155 East 44th Street, New York, New York 10017.)

National Fire Protection Association (NFPA)

Standard No. 70 - National Electrical Code

(Application for copies should be addressed to the National Fire Protection Association, 60 Batterymarch Street, Boston, Massachusetts 02110.)

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Underwriters' Laboratories, (UL)

Electrical Construction Materials List

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, Ill. 60711; Walt Whitman Road, Melville, N. Y. 11749; or 1655 Scott Boulevard, Santa Clara, Calif. 95050.)

3. REQUIREMENTS

* 3.1 First article.- When specified (see 6.2), the supplier shall furnish a sample for first article inspection and approval (see 4.2 and 6.5).

3.2 Standard product.- The laundry bundle conveyors covered by this specification shall be a regular commercial product of the manufacturer, except for any changes required in order to comply with the requirements of this specification.

* 3.3 Code and standards.-

3.3.1 NFPA.- The installation of the electrical components of the conveyor shall conform to the requirements of NFPA Standard No. 70.

3.3.2 UL.- The electrical materials and components shall comply with UL standards, as applicable.

3.3.3 NEMA.- Motor controllers shall conform to performance requirements of NEMA Publication No. ICS-1970.

3.3.4 Certification.- Prior to approval of the first article, or if none is submitted, prior to approval of the first shipment, the supplier shall submit satisfactory evidence to the contracting officer or his authorized representative, that the electrical materials and components he proposes to furnish meet the requirements specified in 3.3.2 and 3.3.3, and that the components are installed as specified in 3.3.1.

3.3.4.1 NFPA.- Acceptable evidence of meeting requirements of 3.3.1 shall be the manufacturer's certified statement that the installation of the electrical components conforms to NFPA Standard No. 70.

3.3.4.2 UL.- Acceptable evidence of meeting requirements of 3.3.2 shall be the UL label, listing mark or listing in UL components list, or certification from an independent testing laboratory acceptable to the contracting officer or his authorized representative that the electrical materials and components conform to applicable UL standards.

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3.3.4.3 NEMA.- Acceptable evidence of meeting the requirements of 3.3.3 shall be the manufacturer's certified statement that motor controllers conform to NEMA Standards Publication No. ICS-1970. A tag or label attached to the controller(s) stating the component conforms to this standard is acceptable evidence.

3.4 Materials.- Materials not definitely specified shall be of the quality normally used by the manufacturer provided the completed item complies with all the provisions of this specification.

* 3.5 Design and construction.- The conveyor shall be designed and constructed to support storage laundry bins, that travel on a rail to a preselected location, automatically within the limits of its travel. The conveyor shall be electrically powered. Overall dimensions of the conveyor, including bins shall be as specified in table I. Each bin shall be provided with shelf supports having a minimum vertical height of 6 inches between supports to receive removable shelves. The conveyor shall be floor mounted and shall support a capacity load as specified in table I. When operationally and functionally tested as specified in 4.4.1 and 4.4.2, the completely assembled conveyor shall show that the assembly of components have:

(a) Complete and proper relationship of moving parts.
(b) Motor bearings that show no evidence of overheating.
(c) Starting and stopping devices operating smoothly and positively.
(d) No structural failures.

TABLE I.- Dimensions

Size	Cubic feet	Overall height (maximum)	Overall length (± 12 inches)	Overall width (± 6 inches)	Min. load	No. of bins
A	175	98 inch	12 ft 6 in	56 inches	1300 lbs	10
B	300	98 inch	20 ft 6 in	56 inches	2100 lbs	18
C	365	98 inch	24 ft 6 in	56 inches	2800 lbs	22
D	480	98 inch	31 ft 6 in	56 inches	3600 lbs	30
E	1/	1/	1/	1/		1/

1/ As specified.

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3.5.1 Frame.- The continuous rail for bin travel shall form a part of the frame, shall be of steel and contain not less than two hangers for each bin. The hangers shall travel on the rail by use of ball bearing rollers with rubber or nylon tread. The bin transporter and bins (see 3.5.5.1) shall contain identifiable numerals or letters designating each bin. The frame base shall be provided with sufficient anchor bolt holes for anchoring the assembled conveyor to the floor to prevent any tipping when the conveyor is carrying an unbalanced load. The base of the frame shall be provided with a guide rail for guiding the bins during travel. Means shall be provided to allow adjustment for leveling the frame.

3.5.2 Bins.- The bins shall be 72 inches high, 23 inches wide, and 18 inches deep, all ± 1 inch, and shall be fabricated from corrosion-resistant steel wire or carbon steel wire, zinc or cadmium plated. The shelves which form the compartments in the bins shall be removable and all shelves shall be pitched downward to the rear. Means shall be provided to fasten and remove the bins from the hangers on the travel rail without special tools. Each bin shall be equipped with not less than two casters or rollers having rubber or nylon treads for riding against the guide rail.

* 3.5.3 Drive.- The conveyor shall be driven by an electric motor or motors through a V-belt and pulleys, or by gears, or roller chain and sprocket. The conveyor shall operate at a rated speed of 50 to 90 feet per minute (FPM) when tested as specified in 4.4.1 and 4.4.2.

3.5.4 Electrical characteristics.- Unless otherwise specified (see 6.2), the conveyor shall operate on a 120 volt, 60-hertz, 1-phase power source. The motor shall conform to CC-M-636 or CC-M-641 as applicable.

* 3.5.5 Controls.- Unless otherwise specified (see 6.2), each conveyor shall be equipped with both manual and automatic electric controls.

3.5.5.1 Automatic control.- The automatic control shall be of the remote type. The control shall have a dial or push-button system with numerals or letters, corresponding with the numerals or letters on the bins or the bin transporter, designating each bin. When the control is dialed or a push-button pressed for any particular numeral or letter, the control shall activate a drive motor switch. The conveyor shall then operate in a direction of shortest travel and accurately stop the bin with the corresponding numeral or letter at an established station.

3.5.5.2 Manual control.- Each conveyor shall be equipped with a pair of foot switches (right and left), which, when stepped on, shall cause the conveyor to revolve in either direction, depending upon the switch pressed. The conveyor shall stop immediately when the switch is released.

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3.6 Finish.- The conveyor shall be finished in accordance with the manufacturer's standard practice.

3.6.1 Application of finish.- The finish applied to the end item shall be continuous, smooth, adherent, without discoloration or foreign material imbedded, and contain no sags, runs, drips, creeps, laps, bubbles, cracks, pours, pits, lumps, flux, or orange peel. No rust, rough grinds or toolmarks shall show through the coating.

3.7 Marking.- Each laundry bundle conveyor shall be marked for identification in accordance with MIL-STD-130.

* 3.8 Electromagnetic compatibility.- When specified (see 6.2), equipment shall be designed and equipped for electromagnetic compatibility in accordance with class IIB of MIL-STD-461 (see 4.4.3).

3.9 Workmanship.- The conveyor shall be free from defects such as fractures, splits, punctures, tears, dents, creases, bows, deteriorations, or malformations. There shall be no sharp edges, burrs, or slivers.

3.9.1 Welding.- Welds shall be continuous, sound, smooth, and free from porosity, cracks, incomplete fusion, and deformation of material. All scale and flux deposits shall be removed from finished welds.

3.9.2 Bolts, nuts, screws, studs, and other threaded fasteners.- Threaded fasteners shall not be driven, stripped, fractured, or loose.

3.9.3 Driving of rivets.- Rivets shall be driven to completely fill the holes, with full size heads neatly finished and in full contact with the surface of the members.

3.9.4 Electrical wiring.- Wiring shall not be cut, abraded, or have excessive insulation stripped, and shall be properly and tightly joined at terminals.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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* 4.2 First article inspection.- When a first article is required (see 6.5), it shall be examined for defects in table II, and specified dimensions, and tested as specified in 4.4.1, 4.4.2, and when specified 4.4.3. The presence of any visual defect, any dimension not within specified requirements, or failure to pass the tests shall be cause for rejection of the first article.

4.3 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Component and material inspection.- In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.3.2 End item inspection.- An inspection lot shall consist of all conveyors of one size offered for inspection at one time. The sample unit shall be one completely fabricated conveyor.

* 4.3.2.1 Visual examination.- Examination shall be made of the end item for defects listed in table II. The inspection level shall be II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE II.- Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Not finished where required		X
	Finish not continuous, smooth, adherent		X
	Contains discolorations, foreign material, sags, runs, drips, creeps, laps, bubbles, cracks, pours, pits, lumps, flux, or orange peel		X
	Rust, rough grinds or tool marks show through		X
	Finish not as specified	X	
Design and construction	Any part missing or not specified type or size	X	
	Splits, tears, dents, creases, deteriorations, or malformed		X
	Any component misplaced, not in proper alignment, or not joined properly	X	

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TABLE II.- Classification of defects (cont'd)

Examine	Defect	Classification Major	Classification Minor
Design and construction (cont'd)	Any component loose		X
	Sharp burr, sliver, or sharp edge	X	
	Any component that is inoperative or will not function as intended	X	
	Any component that requires abnormal force to operate	X	
Welding	Missing, incomplete, burn holes, fractured, or not fused	X	
	Slag inclusion, undercut, not smooth and uniform, or deposits of scale and flux not removed		X
Rivets	Not in full contact with surfaces connected		X
Bolts, nuts, screws, and other threaded fasteners	Missing, broken, stripped, or fractured	X	
	Threaded fastener loose		X
Electrical assembly	Wiring cut, abraded		X
	Adequate slack not provided for wiring or excessive insulation stripped from wiring	X	
Identification marking	Missing, incomplete, not legible		X

4.3.2.2 Dimensional examination.- Examination shall be made of the end item for compliance with dimensional requirements. Any deviation from the specified requirements shall be classified a defect. The inspection level shall be S-2 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

4.3.2.3 Testing of end item.- Unless otherwise specified (see 6.3), when a first article is not required, the initial production unit shall be tested as specified in 4.4.2 and when applicable, 4.4.3. Any nonconformance shall be cause for rejection of the initial article of production. Each unit of production shall be tested as specified in 4.4.1. Any nonconformance shall be cause for rejection of the unit.

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4.3.3 Code and standards compliance.- Proof of compliance with the requirements of 3.3 shall be submitted to the Government representative.

4.3.4 Examination of preparation for delivery.- An examination shall be made to determine that preservation and packaging, packing, and marking as required by section 5 of this specification are complied with.

4.4 Tests.-

4.4.1 Operational testing.- Each conveyor furnished under this specification shall be connected to power and operationally tested, without loading bins. In the course of testing for operational requirements specified in 3.5, determination shall be made to establish compliance with all performance requirements applicable to the fully assembled item except those that require functional testing such as operating the conveyor for load test. The test shall be maintained for a minimum of 30 minutes to determine compliance with 3.5. Any nonconformance with operational requirements of 3.5 and 3.5.3 shall constitute failure of this test.

4.4.2 Functional test.- The conveyor shall be tested by making connection to electric power and loading all shelves uniformly to capacity (see table I). The loaded conveyor shall be continuously revolved by its drive for not less than 15 minutes to establish conformance with the requirements of 3.5. Any nonconformance with functional requirements of 3.5 and 3.5.3 shall constitute failure of this test.

* 4.4.3 Electromagnetic compatibility.- When electromagnetic compatibility is required, the preproduction sample or initial unit of production, as applicable, shall be tested by the supplier in accordance with test methods CE03 and RE02 of MIL-STD-462. The Government reserves the right to witness tests performed by the supplier or an independent testing agency. The supplier shall furnish the contracting officer a written certification that the Interference Control Plan, the EMI/EMC Test Plan, the electromagnetic compatibility test report and the requirements meet MIL-STD-461.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging, packing, and marking.- Preservation and packaging, packing, and marking for level A, B, or C shall be in accordance with MIL-L-3153 (see 6.2).

6. NOTES

6.1 Intended use.- The laundry bundle conveyor covered by this specification is intended for use in laundry call offices for storing and automatically conveying laundry bundles to an established station for ready dispersement by the operator.

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6.2 Ordering data.- Procurement documents should specify the following, as applicable:

(a) Title, number, and date of this specification.
(b) Size of conveyor required (see 1.2).
(c) When a first article is required (see 3.1, 4.2 and 6.5).
(d) Electrical characteristics (voltage, phase and frequency) when other than specified (see 3.5.4).
(e) When two types of controls are not required (see 3.5.5).
(f) When electromagnetic compatibility is required (see 3.8).
(g) Selection of applicable levels of preservation and packaging, and packing (see 5.1).

* 6.2.1 Contract data requirements.- Any requirements for equipment manuals for the item covered by this specification should be included in DD Form 1423 Contract Data Requirements List and cited in the contract.

6.2.2 Instruction manual.- When procurement is for the Air Force, one instruction manual giving operating, installation, maintenance instructions including pictures or illustrations as necessary, dimensions, floor plans of machine, electrical data, and connection diagrams for all utilities, shall be furnished and packed with each conveyor.

6.2.3 Parts manual.- When procurement is for the Air Force, one copy of the manufacturer's illustrated repair parts manual shall be furnished with each conveyor. The manual shall include the supplier's part number, the part number assigned by the actual manufacturer, name of actual manufacturer, and description of parts. Each specific repair part shall be assigned a specific part number, and part numbers shall not be duplicated by a given manufacturer regardless of model number or end item application, unless an interchangeable factor exists; then, in this case, the manufacturer shall assign identical part numbers to identical parts. Figure, drawing, item, plate, or casting numbers may be used for cross reference only and shall not be used in lieu of part numbers. The requirements specified in this paragraph may be included in the instruction manual specified in 6.2.2.

6.3 Functional testing.- The functional test (4.4.2), required in 4.3.2.3 for the first unit of production, may be waived by the contracting officer, if this test has been performed previously on an identical conveyor and the supplier provides certified test data indicating compliance with the requirement.

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6.4 Successful commercial operation.- Invitation for bids should specify that no item of equipment will be acceptable unless the manufacturer has had equipment of approximately the same design as that specified, operating successfully in a commercial or institutional laundry or laundry call office for at least 1 year. Equipment installed for test purposes in a manufacturer's plant or laboratory shall not come within the category of successful commercial operation.

* 6.5 First article.- When a first article is required, it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The first article should consist of one conveyor of the size specified. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

* 6.6 The margins of this specification are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Air Force - 84

Preparing activity:

Army - GL

Project No. 3510-0192

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION CONVEYOR, LAUNDRY, STORAGE BINS, ELECTRIC POWERED, AUTO SELECTIVE CONTROL MIL-C-43532A

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 61 1426

REPLACES EDITION OF 1 OCT 54 WHICH MAY BE USED.

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